

Benchmarking AI and Operational Subseasonal Forecast Systems for Indian Winter Precipitation Prediction During JFM 2026

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Abstract

Subseasonal-to-seasonal (S2S) forecasts at lead times of two to six weeks support planning decisions in agriculture, water resources, and disaster preparedness, yet precipitation skill at these leads remains limited, particularly over tropical and subtropical land regions. Machine-learning (ML) forecast systems trained on reanalysis data now match or exceed operational numerical weather prediction at medium range, but their subseasonal precipitation skill over regional domains such as India is largely undocumented, especially in a real-time setting. Here we benchmark two AI-based S2S systems—Spire AI-S2S and FuXi-S2S—against two operational dynamical systems—the ECMWF extended-range ensemble and the NCEP Climate Forecast System version 2 (CFSv2)—for weekly precipitation prediction over India during January–March (JFM) 2026. Ensemble-mean forecasts from 13 weekly initializations (1 January–26 March 2026) are verified against ERA5 at weekly leads of 1–4 weeks using the anomaly correlation coefficient (ACC), root-mean-square error (RMSE), and mean bias, computed over the Indian land domain. [Headline result sentence, e.g.: “Week-1 ACC ranges from X.XX to X.XX across systems and declines to X.XX–X.XX by week 3, with SYSTEM retaining the highest skill at all leads.”] [One sentence on the spatial structure of skill, e.g., concentration over northwest India and the Western Himalaya in association with western disturbance activity.] [One sentence on systematic biases, e.g., sign and amplitude of the mean precipitation bias in the AI versus dynamical systems.] The results provide an early, real-time-oriented assessment of AI-based S2S precipitation forecasts for the Indian winter season and establish a verification template that can be extended to additional systems, variables, and seasons as they become available.

Keywords: subseasonal-to-seasonal prediction; machine-learning weather forecasting; precipitation verification; India; western disturbances; ensemble forecasting

1 Introduction

Subseasonal-to-seasonal (S2S) prediction, covering lead times from roughly two weeks to two months, bridges the gap between medium-range weather forecasting and seasonal climate outlooks (Vitart et al., 2017; National Academies, 2016). Forecasts at these leads are increasingly demanded for agricultural planning, reservoir operations, energy load management, and disaster preparedness (White et al., 2017), yet they remain among the most difficult products to produce skillfully: the atmosphere has largely lost its initial-condition memory, while slowly varying boundary forcings have not yet come to dominate (Mariotti et al., 2020). Documented

sources of subseasonal predictability include the Madden–Julian oscillation (MJO; [Zhang, 2005](#)), stratosphere–troposphere coupling, soil-moisture and snow anomalies, and the slowly evolving state of the El Niño–Southern Oscillation (ENSO) ([Mariotti et al., 2020](#); [Domeisen et al., 2022](#)). Precipitation, the variable of greatest practical value to many users, is also the variable for which S2S skill degrades most rapidly: hindcast-based assessments show that anomaly correlations over most tropical and subtropical land regions fall to modest values beyond week 2 ([de Andrade et al., 2019](#); [Pegion et al., 2019](#)).

Over India, the January–March (JFM) season—boreal winter transitioning into early spring—has received far less attention in the S2S literature than the summer monsoon, despite its societal importance. Winter precipitation over northern India and the Western Himalaya is produced chiefly by western disturbances (WDs), extratropical synoptic-scale systems embedded in the subtropical westerly jet ([Dimri et al., 2015](#); [Hunt et al., 2018](#)). WD-associated precipitation sustains the rabi (winter) cropping season, including the wheat belt of the Indo-Gangetic Plain, and builds the Himalayan snowpack that feeds the Indus and Ganges river systems during the subsequent melt season ([Dimri et al., 2015](#)). Southern peninsular India can additionally receive residual northeast-monsoon rainfall in early January, while anomalously wet or dry winters modulate fog occurrence, cold-wave conditions, and pre-monsoon water availability. Skillful multi-week precipitation outlooks for JFM therefore carry substantial value for Indian agriculture and water management, and extended-range forecasting for the region is an operational priority of the India Meteorological Department (IMD) and the Ministry of Earth Sciences ([Rao et al., 2019](#); [Pattanaik et al., 2019](#)).

In parallel, weather prediction is being reshaped by machine-learning (ML) forecast systems trained on reanalysis data. Deterministic data-driven models such as FourCastNet, Pangu-Weather, and GraphCast now match or exceed operational numerical weather prediction (NWP) on standard medium-range headline scores ([Pathak et al., 2022](#); [Bi et al., 2023](#); [Lam et al., 2023](#); [Ben Bouallègue et al., 2024](#); [Rasp et al., 2024](#)); probabilistic ML systems have followed ([Price et al., 2025](#)); hybrid approaches couple learned parameterizations to dynamical cores ([Kochkov et al., 2024](#)); and operational centers have begun running data-driven systems alongside their NWP suites ([Lang et al., 2024](#)). Extension of these methods to the subseasonal range is more recent: FuXi-S2S, an ML system producing 42-day ensemble forecasts, has been reported to outperform conventional S2S models on several global metrics ([Chen et al., 2024](#)), and commercial providers now issue AI-based S2S products ([Spire Global, 2026](#)). Published evaluations of ML S2S systems have, however, emphasized global or large-scale diagnostics—for example, MJO prediction and midlatitude circulation indices—while regional precipitation skill over densely populated, hydrologically sensitive domains such as India remains largely undocumented.

A further gap concerns the evaluation setting. Most S2S skill estimates derive from multi-year hindcasts ([de Andrade et al., 2019](#); [Pegion et al., 2019](#)), which provide robust statistics but may not reflect the configuration of a system as actually operated in a given season—particularly for rapidly evolving AI models. Real-time (or quasi-real-time) verification of a single season offers a complementary, if statistically noisier, view: it tests the systems as users experienced them, with operational initial conditions and operational model versions. In this study we adopt this real-time perspective for JFM 2026, benchmarking two AI-based systems—Spire AI-S2S and FuXi-S2S—against two established operational dynamical systems—the ECMWF extended-range ensemble and NCEP CFSv2—for weekly precipitation over India.

Specifically, we address four questions:

1. How does weekly precipitation forecast skill over India decay with lead time (weeks 1–4) in each system during JFM 2026?
2. Are the AI-based systems competitive with the operational dynamical systems, and at which leads?
3. How is skill organized spatially, and is its structure physically interpretable in terms of

WD activity and regional precipitation regimes?

4. What systematic biases characterize each system, and do these differ between the AI-based and dynamical models?

We verify ensemble-mean forecasts from 13 weekly initializations against ERA5 using the anomaly correlation coefficient (ACC), root-mean-square error (RMSE), and mean bias. The remainder of the paper is organized as follows. Section 2 describes the verification data and study domain, Section 3 the processing methodology, Section 4 the forecast systems, and Section 5 the verification framework. Results are presented in Section 6 and discussed in Section 7; limitations, future work, and conclusions follow in Sections 8–10.

2 Data

2.1 Verification reference: ERA5

We use the fifth-generation ECMWF reanalysis (ERA5; [Hersbach et al., 2020](#)) as the verification reference. Hourly total precipitation (`tp`) on the native $0.25^\circ \times 0.25^\circ$ grid is accumulated to daily totals (00–00 UTC) and subsequently to the weekly windows defined in Section 3.1. ERA5 precipitation is a short-range forecast product of the IFS rather than a direct observational analysis, and its quality over the Indian region varies spatially, with the largest uncertainties over the Western Himalaya, where gauge density is low and orographic gradients are strong. We therefore treat ERA5 as a dynamically consistent, temporally complete reference suitable for anomaly-based verification, and we discuss the implications of this choice—together with a planned cross-check against the IMD 0.25° gauge-based rainfall analysis ([Pai et al., 2014](#))—in Sections 8 and 9. [If a second high-resolution verification dataset (e.g., IMD gridded rainfall or IMERG) is added to the final analysis, document it here and in Table 1.] ERA5 data through [22 April 2026] are required to close the final week-4 verification window (Table 1).

2.2 Study domain

The study domain covers the Indian region, [6.5° – 38.5° N, 66.5° – 100° E (suggested; finalize)], with all aggregate skill metrics computed over land points within [the Indian land mask / the domain land mask — specify] (Fig. 1). Area weighting by the cosine of latitude is applied throughout. [If sub-regions are analyzed (e.g., northwest India, the Western Himalaya, the Indo-Gangetic Plain, peninsular India), define them here and overlay their outlines in Fig. 1.]

2.3 Forecast data

Weekly initialized forecasts are collected from the four systems summarized in Table 2 for the 13 initialization dates listed in Section 3.1. ECMWF and NCEP forecasts are obtained from [the WWRP/WCRP S2S database ([Vitart et al., 2017](#)) / ECMWF MARS / NOAA archives — specify source and any embargo handling]. Spire AI-S2S forecasts are provided by [Spire Global under a research arrangement — specify]. FuXi-S2S forecasts are [obtained from a public archive / generated locally from the publicly released model weights, initialized from ERA5(-like) analyses at 00 UTC on each start date — specify, including any differences from the configuration documented by [Chen et al. \(2024\)](#)].

3 Methodology

3.1 Initialization dates and weekly lead windows

Thirteen initializations at weekly intervals span the JFM 2026 season: 1, 8, 15, 22, and 29 January; 5, 12, 19, and 26 February; and 5, 12, 19, and 26 March 2026 (all at [00 UTC — confirm

FIGURE PLACEHOLDER

Figure 1 — Map of the India study domain. Suggested content: topographic shading; thick outline of the land verification mask; dashed boxes for optional sub-regions; inset timeline of the 13 initialization dates. File: figs/fig01_domain.pdf

Figure 1: The India study domain ($[6.5^{\circ}\text{--}38.5^{\circ}\text{N}, 66.5^{\circ}\text{--}100^{\circ}\text{E}]$). Shading shows [topographic elevation]; the thick outline marks the land verification mask used for all area-aggregated scores, and dashed boxes denote [optional sub-regions used in the regional analysis]. [Optionally include an inset showing the 13 weekly initialization dates spanning 1 January–26 March 2026.]

Table 1: Verification dataset and protocol summary.

Property	Specification
Reference dataset	ERA5 (Hersbach et al., 2020)
Variable	Total precipitation (tp)
Native resolution	$0.25^{\circ} \times 0.25^{\circ}$, hourly
Verification grid	$[1.5^{\circ}]$ common grid (conservative remapping)
Temporal aggregation	Hourly $\rightarrow$ daily (00–00 UTC) $\rightarrow$ weekly
Verification period	1 January – [22 April] 2026
Observed climatology	[1991–2020], daily, [smoothing method]
Spatial mask	Indian land points, $\cos \varphi$ area weights
Secondary reference	[IMD 0.25° gauge analysis (Pai et al., 2014) — planned]

per system]). Forecast precipitation is aggregated into nonoverlapping weekly accumulation windows: week 1 corresponds to forecast days 1–7, week 2 to days 8–14, week 3 to days 15–21, and week 4 to days 22–28, where day 1 denotes the first 24 h after initialization. The verifying periods collectively span 1 January–[22 April] 2026. [If any system provides output at a different time resolution or initialization time (e.g., daily-mean rates rather than accumulations, or non-00 UTC cycles), document the alignment and conversion here.] Where forecast horizons permit (Table 2), weeks 5–6 may be added in a future revision.

3.2 Common grid

All forecasts and the ERA5 reference are remapped to a common $[1.5^{\circ}]$ (suggested, matching the S2S archive grid; alternatives: 1.0° or 0.25°) latitude–longitude grid prior to verification. First-order conservative remapping is used for precipitation to preserve area-integrated totals. [If results are sensitive to the verification grid, report a sensitivity test (e.g., repeat Table 3 on the 0.25° grid).]

Table 2: Forecast systems evaluated in this study. All specifications refer to the configurations operational (or executed) during JFM 2026. [Cells in red must be confirmed against provider documentation before submission.]

System	Type	Resolution	Members	Horizon	Reforecasts
ECMWF ext-range	Dynamical, coupled	[Tco319 (≈ 36 km)]	[101]	46 d	[on-the-fly, 20 yr]
NCEP CFSv2	Dynamical, coupled	[T126 (≈ 100 km)]	[N (lagged)]	45 d	[1999–2010]
Spire AI-S2S	AI (data-driven)	[TBD]	[TBD]	[TBD]	[TBD]
FuXi-S2S	AI (data-driven)	[1.5° — confirm]	51	42 d	[period — confirm]

3.3 Climatologies and anomaly definition

Observed anomalies are computed by removing a daily ERA5 climatology for [1991–2020], smoothed with [a 31-day running mean / the first four annual harmonics — state choice], and aggregated to the same weekly windows. Forecast anomalies are computed, wherever possible, with respect to each system’s own lead-dependent model climatology derived from its hindcasts (reforecasts), which removes the lead-dependent mean drift in the standard S2S manner (Vitart et al., 2017): ECMWF [on-the-fly reforecasts spanning the preceding 20 years, N members per start date — confirm]; NCEP CFSv2 [1999–2010 reforecast climatology — confirm]; FuXi-S2S [reforecast period and configuration — specify]; Spire AI-S2S [reforecast availability — to be determined]. For any system lacking an adequate reforecast set, forecast anomalies are instead defined [relative to the ERA5 climatology after removal of the system’s JFM 2026 in-sample mean bias / relative to the raw ERA5 climatology — state the final choice per system]. We note that anomalies based on in-sample bias correction can yield modestly optimistic ACC values; [describe any safeguard adopted, e.g., a leave-one-out correction across the 13 start dates].

3.4 Ensemble treatment

Verification targets the ensemble mean of each system; member counts are given in Table 2. Ensemble means computed from unequal ensemble sizes are not strictly comparable, because larger ensembles damp unpredictable variance more effectively. [Optionally report a fixed-size subsampled comparison (e.g., the first N members of every system, with N set by the smallest ensemble) as a sensitivity test.] Probabilistic verification is deferred to future work (Section 9).

4 Forecast Models

Table 2 summarizes the four systems evaluated. [Verify every entry in Table 2 against provider documentation for the versions operational during JFM 2026.]

4.1 ECMWF extended-range ensemble

The ECMWF extended-range configuration (Vitart, 2014) provides ensemble forecasts to day 46. During the study period the system ran [daily at 00 UTC with 101 members at Tco319 (≈ 36 km) resolution under IFS cycle [confirm cycle(s) operational during JFM 2026, and whether any upgrade occurred within the period]]. On-the-fly reforecasts supply the lead-dependent model climatology used for anomaly computation (Section 3.3).

4.2 NCEP CFSv2

The NCEP Climate Forecast System version 2 is a fully coupled atmosphere–ocean–land system, operational since 2011 (Saha et al., 2014), with a [T126 (≈ 100 km), 64-level] atmosphere. [Specify the runs used here: e.g., the four daily 00/06/12/18 UTC cycles aggregated into a lagged

ensemble of N members per weekly initialization, taking the cycles nearest each nominal start date.] The [1999–2010] CFSv2 reforecast set provides the model climatology.

4.3 Spire AI-S2S

Spire AI-S2S is a commercially produced, ML-based subseasonal forecasting system (Spire Global, 2026). [Insert the system description as documented by the provider: model architecture and training data; initialization source; output resolution and time step; ensemble generation method and size; forecast horizon; reforecast availability.] Forecasts used in this study cover [the JFM 2026 window at weekly initialization frequency — confirm delivery schedule].

4.4 FuXi-S2S

FuXi-S2S is an ML subseasonal forecasting system based on the FuXi architecture, producing daily-mean global forecasts to day 42 with a 51-member ensemble generated through perturbations [of the initial conditions (Perlin-noise based) and/or model perturbations — confirm the configuration used here] (Chen et al., 2024). [Specify: the source of the 2026 initial conditions; whether the publicly released weights were used unmodified; the output grid; and the conversion of daily-mean precipitation rates to weekly accumulations.] [If FuXi-S2S is initialized from ERA5 while the operational systems use their own analyses, state this explicitly; the implications are discussed in Section 8.]

5 Verification Framework

Let $f'_{i,t,\ell}$ denote the ensemble-mean forecast anomaly at land grid point i , initialization t ($t = 1, \dots, T$; $T = 13$), and weekly lead $\ell \in \{1, 2, 3, 4\}$, and let $o'_{i,t,\ell}$ denote the corresponding ERA5 anomaly for the same verifying week. Area weights are $w_i = \cos \varphi_i$, with sums taken over the land mask of Fig. 1.

5.1 Metrics

The anomaly correlation coefficient at lead ℓ is the centered spatial (pattern) correlation computed per initialization and averaged over the season,

$$\text{ACC}(\ell) = \frac{1}{T} \sum_{t=1}^T \frac{\sum_i w_i f'_{i,t,\ell} o'_{i,t,\ell}}{\sqrt{\sum_i w_i f'^2_{i,t,\ell}} \sqrt{\sum_i w_i o'^2_{i,t,\ell}}}. \quad (1)$$

[State whether the average in Eq. (1) is taken on raw correlations or on Fisher- z transforms.] The root-mean-square error is

$$\text{RMSE}(\ell) = \frac{1}{T} \sum_{t=1}^T \sqrt{\frac{\sum_i w_i (f_{i,t,\ell} - o_{i,t,\ell})^2}{\sum_i w_i}}, \quad (2)$$

evaluated on [full weekly accumulations in mm week^{-1} (suggested) and/or on anomalies — state choice], and the mean bias is

$$\text{Bias}(\ell) = \frac{1}{T} \sum_{t=1}^T \frac{\sum_i w_i (f_{i,t,\ell} - o_{i,t,\ell})}{\sum_i w_i}, \quad (3)$$

in mm week^{-1} . Importantly, mean bias is computed from *raw* (uncorrected) weekly accumulations: for systems whose anomalies are defined against their own reforecast climatology

(Section 3.3), the anomaly-space bias vanishes by construction, so Eq. (3) is the informative quantity about each system’s native precipitation climate over India.

For the spatial skill maps (Fig. 4), the temporal ACC at each grid point is the correlation, across the $T = 13$ initializations, between forecast and observed weekly anomalies. With so few samples these maps are necessarily noisy; they are interpreted qualitatively, with significance assessed by [a two-sided bootstrap / t -test at the 5% level, indicated by stippling].

5.2 Reference forecasts

Two reference forecasts contextualize all scores: (i) climatology (zero anomaly), for which the ACC of Eq. (1) is undefined and the anomaly RMSE equals the observed anomaly standard deviation; and (ii) persistence, defined as the observed anomaly of the week immediately preceding initialization, applied at all leads. [Confirm inclusion of both baselines in Figs. 2 and 3; recommended.]

5.3 Uncertainty estimation

Confidence intervals for all aggregate scores are estimated by bootstrap resampling of the 13 initialization dates ([1000] resamples; percentile [95%] intervals). Differences between systems are assessed by paired resampling of score differences [(describe procedure)]. Given $T = 13$, only large differences are expected to be distinguishable, and all statements of relative skill in Section 6 should be read with this caveat.

5.4 Interpreting deterministic scores for smooth forecasts

Ensemble means—and, by construction, many ML forecasts—are smoother than individual realizations of the atmosphere. Deterministic scores such as RMSE therefore structurally reward low-variance forecasts (the “double penalty” effect; Wilks, 2019; Ben Bouallègue et al., 2024). To make this effect visible we additionally report [the ratio of forecast-anomaly to observed-anomaly spatial standard deviation at each lead (suggested; inexpensive and informative)] and interpret RMSE alongside ACC and bias rather than in isolation (Murphy, 1988).

6 Results

[NOTE TO AUTHORS: Every numerical value, ranking, and descriptive statement in this section is a placeholder template. Replace with computed results and delete this notice before submission.]

6.1 Lead-time-dependent skill

Figure 2 shows the domain-mean pattern ACC of weekly precipitation anomalies as a function of forecast lead for all four systems, together with the persistence baseline. Averaged over the 13 initializations, ECMWF attains an ACC of [X.XX] at week 1, declining to [X.XX] at week 2, [X.XX] at week 3, and [X.XX] at week 4. The corresponding values are [X.XX], [X.XX], [X.XX], and [X.XX] for NCEP CFSv2; [X.XX], [X.XX], [X.XX], and [X.XX] for Spire AI-S2S; and [X.XX], [X.XX], [X.XX], and [X.XX] for FuXi-S2S. Taking $\text{ACC} \geq 0.5$ as an indicative threshold of useful deterministic skill, [state which systems exceed it at which leads]. Bootstrap confidence intervals [overlap / separate] the systems at weeks [X–Y], indicating that [summarize].

The anomaly RMSE (Fig. 3) exhibits [describe behavior: growth with lead, saturation relative to the climatology baseline, and any reordering of systems relative to the ACC ranking]. We note that RMSE comparisons across systems with different effective smoothness must be interpreted in light of Section 5.4; the forecast-to-observed anomaly standard deviation ratio at

FIGURE PLACEHOLDER

Figure 2 placeholder: domain-mean weekly ACC vs. lead (weeks 1–4), one line per system + persistence baseline, with bootstrap 95% CIs.

Figure 2: Domain-mean anomaly correlation coefficient (ACC) of weekly precipitation anomalies over the India domain as a function of forecast lead (weeks 1–4), averaged over the 13 JFM 2026 initializations, for ECMWF, NCEP CFSv2, Spire AI-S2S, and FuXi-S2S. The persistence baseline is shown in gray. Shading denotes bootstrap [95%] confidence intervals. [Replace with computed figure.]

Table 3: Domain-mean ACC of weekly precipitation anomalies over the India domain, averaged over the 13 JFM 2026 initializations. [Bold the best system per lead. Brackets: bootstrap 95% CIs.]

System	Week 1	Week 2	Week 3	Week 4
ECMWF	[X.XX]	[X.XX]	[X.XX]	[X.XX]
NCEP CFSv2	[X.XX]	[X.XX]	[X.XX]	[X.XX]
Spire AI-S2S	[X.XX]	[X.XX]	[X.XX]	[X.XX]
FuXi-S2S	[X.XX]	[X.XX]	[X.XX]	[X.XX]
Persistence	[X.XX]	[X.XX]	[X.XX]	[X.XX]

each lead is [X.XX] (ECMWF), [X.XX] (NCEP CFSv2), [X.XX] (Spire AI-S2S), and [X.XX] (FuXi-S2S) at week 3, indicating [degree of smoothing].

Tables 3 and 4 summarize the scores.

6.2 Spatial distribution of skill

Figure 4 maps the temporal ACC (computed across the 13 initializations at each grid point; Section 5) for each system and lead. [Describe the resulting spatial structure. Plausible hypotheses to test against the computed maps: (i) higher week-1–2 skill along the western-disturbance corridor of northwest India and the western Himalaya, where synoptic forcing dominates JFM precipitation; (ii) lower and noisier skill over the weakly forced peninsular interior and northeast; (iii) faster skill decay with lead in regions of convective, small-scale precipitation. Do not assert any of these without inspection of the maps; with $n = 13$ samples per grid point the maps are noisy and stippling for $[p < 0.05]$ significance is recommended.]

6.3 Bias analysis

Because anomalies referenced to each system’s own reforecast climatology are unbiased by construction, mean bias is evaluated on raw weekly accumulations (Section 5). Figure 5 shows the domain-mean bias as a function of lead, and Table 5 lists the values. [Describe sign, magnitude, lead dependence, and spatial pattern of each system’s bias; note in particular whether the

FIGURE PLACEHOLDER

Figure 3 placeholder: domain-mean weekly anomaly RMSE vs. lead, one line per system + climatology and persistence baselines.

Figure 3: As in Fig. 2, but for the root-mean-square error (RMSE) of weekly precipitation anomalies (mm week^{-1}). The climatology baseline (observed anomaly standard deviation) and persistence baseline are shown for reference. [Replace with computed figure.]

Table 4: As in Table 3, but for anomaly RMSE (mm week^{-1}). The climatology row equals the domain-mean observed weekly anomaly standard deviation.

System	Week 1	Week 2	Week 3	Week 4
ECMWF	[X.XX]	[X.XX]	[X.XX]	[X.XX]
NCEP CFSv2	[X.XX]	[X.XX]	[X.XX]	[X.XX]
Spire AI-S2S	[X.XX]	[X.XX]	[X.XX]	[X.XX]
FuXi-S2S	[X.XX]	[X.XX]	[X.XX]	[X.XX]
Persistence	[X.XX]	[X.XX]	[X.XX]	[X.XX]
Climatology	[X.XX]	[X.XX]	[X.XX]	[X.XX]

AI systems exhibit the drizzle / light-precipitation overprediction and heavy-tail underprediction commonly reported for ML forecasts, and whether the dynamical systems show orographic precipitation biases along the Himalaya.]

6.4 Case study

To complement the aggregate statistics, we examine [one high-impact JFM 2026 precipitation episode — e.g., the strongest western-disturbance event of the season over northwest India, or the wettest all-India week — selected objectively as the week with the largest domain-mean observed positive anomaly]. Figure 6 compares the ERA5 weekly anomaly with the forecasts of all four systems initialized [1, 2, and 3] weeks before the event. [Describe which systems captured the occurrence, amplitude, and placement of the anomaly at which leads.]

6.5 Synthesis

Figure 7 condenses the verification into a scorecard of [ACC, RMSE skill score relative to climatology, absolute bias, and anomaly-variance ratio] for each system and lead. [Summarize the overall ranking and the leads/metrics for which AI systems are competitive with, superior to, or inferior to the operational dynamical systems. Avoid overstatement: with one season and 13 initializations, rankings are indicative rather than definitive.]

FIGURE PLACEHOLDER

Figure 5 placeholder: matrix of maps, rows = systems (ECMWF, NCEP CFSv2, Spire AI-S2S, FuXi-S2S), columns = leads (weeks 1–4), shading = temporal ACC, stippling = significance.

Figure 4: Temporal anomaly correlation of weekly precipitation anomalies at each grid point over the 13 JFM 2026 initializations, for each forecast system (rows) and lead week (columns). Stippling indicates correlations significant at the [5%] level [(method)]. [Replace with computed figure.]

Table 5: Domain-mean bias of raw weekly precipitation accumulations (mm week^{-1}), averaged over the 13 JFM 2026 initializations.

System	Week 1	Week 2	Week 3	Week 4
ECMWF	[X.XX]	[X.XX]	[X.XX]	[X.XX]
NCEP CFSv2	[X.XX]	[X.XX]	[X.XX]	[X.XX]
Spire AI-S2S	[X.XX]	[X.XX]	[X.XX]	[X.XX]
FuXi-S2S	[X.XX]	[X.XX]	[X.XX]	[X.XX]

7 Discussion

[Rewrite this section once results are final; the paragraphs below are structured prompts, not conclusions.]

First, how do the JFM 2026 scores compare with the hindcast-based expectations in the literature? Multi-model S2S hindcast assessments report useful deterministic precipitation skill over South Asia typically confined to weeks 1–2, with regime- and teleconnection-dependent windows of opportunity at longer leads (de Andrade et al., 2019; White et al., 2017; Mariotti et al., 2020). [State whether the real-time 2026 scores fall within, above, or below those hindcast ranges, and discuss possible reasons (small sample, season-specific predictability, system upgrades).]

Second, the deterministic framing structurally favors smooth forecasts. If the AI systems verify well on RMSE but show reduced anomaly variance (Section 5.4), part of their advantage reflects the double-penalty effect rather than superior placement of precipitation features (Murphy, 1988; Ben Bouallègue et al., 2024; Rasp et al., 2024). [Quantify using the variance ratios and discuss.]

FIGURE PLACEHOLDER

Figure 4 placeholder: domain-mean bias of raw weekly precipitation vs. lead, one line per system, zero line marked.

Figure 5: Domain-mean bias of raw weekly precipitation accumulations (forecast minus ERA5; mm week^{-1}) as a function of lead, averaged over the 13 JFM 2026 initializations. [Replace with computed figure.]

Third, the comparison conflates several factors: ensemble size ([101] vs. [16] vs. [N] vs. [51] members), initialization analyses of differing quality, native resolution, and climatology correction strategies (Section 3.3). Differences in scores should therefore be attributed to “systems as operated,” not to modeling paradigms in the abstract.

Fourth, the large-scale state of JFM 2026 conditions the attainable predictability. [Describe the concurrent ENSO state, MJO activity, and any stratospheric events during JFM 2026, citing monitoring sources, and discuss qualitatively how these may have opened or closed windows of forecast opportunity] (Mariotti et al., 2020; Domeisen et al., 2022; Wheeler and Hendon, 2004; Zhang, 2005).

Finally, the operational context matters for India specifically: winter precipitation governs rabi-season agriculture, Himalayan snowpack, and water resources, and India’s operational extended-range activities provide a natural consumer for such benchmarks (Rao et al., 2019; Pattanaik et al., 2019; Dimri et al., 2015; Hunt et al., 2018). [Discuss practical implications of the findings for these applications.]

8 Limitations

Several limitations qualify the conclusions. The evaluation covers a single season with $T = 13$ initializations, so sampling uncertainty is large, score differences are often statistically indistinguishable, and the results characterize JFM 2026 rather than climatological skill. ERA5 is treated as truth, yet its precipitation is a model product with documented deficiencies over complex terrain, including the Himalaya and orographic northwest India, and verification against gauge-based analyses can alter both magnitudes and rankings (Hersbach et al., 2020; Pai et al., 2014). Only ensemble means are verified, with unequal ensemble sizes across systems, which advantages larger ensembles at long leads and ignores probabilistic information entirely. The anomaly-referencing strategy is heterogeneous — model climatologies derive from reforecast sets of differing length and configuration, and any in-sample correction applied where reforecasts are unavailable optimistically biases the corrected system (Section 3.3). Operational systems may have undergone configuration changes within or shortly before the verification period, so the systems evaluated are snapshots. Deterministic metrics on weekly means mask timing errors within the week and say nothing about sharpness or reliability. Finally, AI systems trained on ERA5 and initialized from (or near) ERA5 analyses enjoy a structural consistency advantage when verified against ERA5 itself, and training-period overlap with the verification season can-

FIGURE PLACEHOLDER

Figure 6 placeholder: top row = ERA5 anomaly for the event week (repeated or single panel); rows below = each system’s forecast anomaly at week-1, week-2, week-3 leads.

Figure 6: Case study of the [date span] precipitation event: observed ERA5 weekly precipitation anomaly (top) and corresponding forecast anomalies from each system at lead times of [1, 2, and 3] weeks. [Replace with computed figure.]

FIGURE PLACEHOLDER

Figure 7 placeholder: scorecard heatmap, rows = system $\times$ metric, columns = lead weeks; or ranking chart.

Figure 7: Summary scorecard of deterministic verification metrics for all systems and lead weeks over the India domain, JFM 2026. [Replace with computed figure.]

not be fully excluded for recently trained systems; these caveats apply with particular force to FuXi-S2S and Spire AI-S2S (Section 4).

9 Future Work

Natural extensions include: (i) adding further systems as access is secured, including FourCastNet-3 [[CITE]], the Deep Learning Earth System Model (Cresswell-Clay et al., 2024), and the IMD Extended Range Forecast System (Pattanaik et al., 2019), the last enabling a direct operational comparison for India; (ii) extending the variable set to 2-m temperature and 500-hPa geopotential height, for which weekly predictability horizons differ markedly from precipitation; (iii) extending verification to weeks 5–6 where forecast ranges permit; (iv) probabilistic verification (CRPS, RPSS, reliability diagrams, ROC) using full ensembles rather than means; (v) tercile categorical verification (above/near/below normal) with Heidke and Gerrity scores, aligning with Indian operational convention (Pattanaik et al., 2019; Wilks, 2019); (vi) stratifica-

tion of skill by MJO phase (Wheeler and Hendon, 2004) and by the concurrent ENSO state to identify windows of opportunity (Mariotti et al., 2020); (vii) multi-winter extension as real-time AI archives lengthen, and companion hindcast verification where reforecasts exist; (viii) cross-verification against IMD gauge-based gridded rainfall and satellite products such as IMERG to test sensitivity to the choice of truth (Pai et al., 2014); and (ix) process-oriented evaluation, in particular objective tracking of western disturbances in forecasts and reanalysis (Hunt et al., 2018; Dimri et al., 2015).

10 Conclusions

We benchmarked two AI-based S2S systems (Spire AI-S2S, FuXi-S2S) against two operational dynamical systems (ECMWF extended range, NCEP CFSv2) for weekly precipitation over India across 13 initializations spanning JFM 2026, verifying ensemble-mean anomalies against ERA5 with ACC, RMSE, and bias at leads of 1–4 weeks. [Insert 3–5 headline findings, e.g.: (1) at week 1, [system] achieved the highest domain-mean ACC ([X.XX]); (2) beyond week [N], no system exceeded ACC = 0.5 over the full domain; (3) AI systems were [competitive / not competitive] with the operational benchmark through week [N], with anomaly variance reduced by [X]%; (4) spatial skill was concentrated in [region]; (5) raw-precipitation biases were [characterize].] Within the limits of a single season, these results [support / qualify] the proposition that current AI S2S systems can provide [competitive] subseasonal precipitation guidance for India, and they establish a reproducible real-time benchmark to be extended in subsequent seasons.

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Data and Code Availability

ERA5 is available from the Copernicus Climate Data Store ([DOI/URL]). ECMWF and NCEP extended-range forecasts are available under the terms of the S2S database and the respective centers’ real-time dissemination policies (Vitart et al., 2017). FuXi-S2S model weights and inference code are available at [URL]. Spire AI-S2S forecasts were provided by Spire Global under [agreement; state access conditions]. All verification and plotting code is available at [GitHub URL / Zenodo DOI].

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